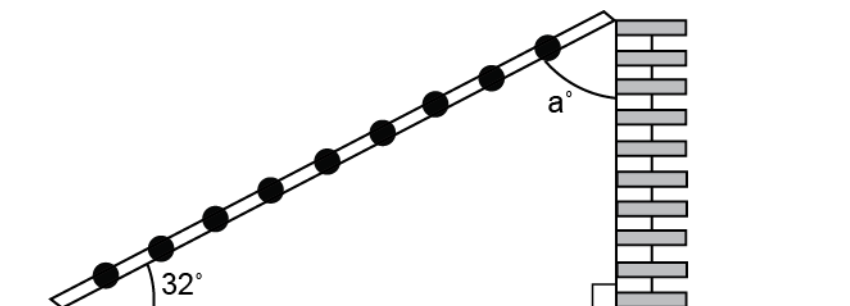


Angle properties of familiar 2D shapes

Teacher resource

Question 1

A ladder leans against a wall making an angle of 32° with the ground.



At what angle does the ladder meet the wall?

23°

☐

48°

☐

58°

☐

78°

☐

Answer: C

Skill: Students use properties of angles in a triangle.

Background information

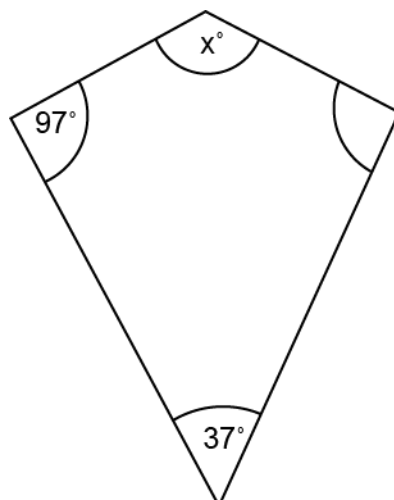
Some secondary students do not understand that a diagram can carry information about angles and dimensions without being drawn to scale and accurately depicting the angles indicated. These students can benefit from experiences in moving between a verbal or written description of a figure, a diagram that shows the relevant information, an accurate construction of the described figure and a scale drawing of the figure.

Additional questions

1. The diagram above may not be drawn to scale. Explain why it is not appropriate to use a protractor to measure angle a to answer the original question.
2. Write equations to express two relationships between angles in the triangle above.

Question 2

A kite has angles as shown below.



What is the size of angle x° ?

46°

☐

97°

☐

113°

☐

129°

☐

Answer: D

Skill: Students use properties of angles in quadrilaterals.

Background information

Some secondary students do not understand that the properties of a class of figures define that class. For example, to show a quadrilateral is a kite, it must have two pairs of adjacent equal sides. This is quite different from the process of identifying a figure as a kite because it 'looks like a kite' and then assuming it must therefore have two pairs of adjacent equal sides because it 'is' a kite.

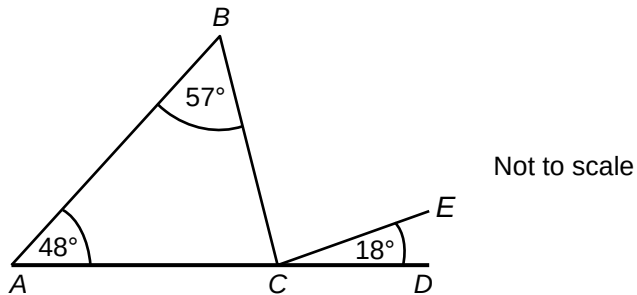
Students will need opportunities to develop an understanding of angle relationships within and the connections between properties of figures. Many students will require physical experiences of constructing figures and then folding, cutting, comparing and re-arranging parts to informally observe relationships between properties before more formal proofs will make sense to them.

Additional questions

1. Do you think the diagram is drawn to scale?
2. Is it possible to construct an accurate drawing of the kite to work out the size of angle x ?

Question 3

In the diagram, ACD is a straight line.



What is the size of angle BCE ?

18°

☐

57°

☐

75°

☐

87°

☐

Answer: D

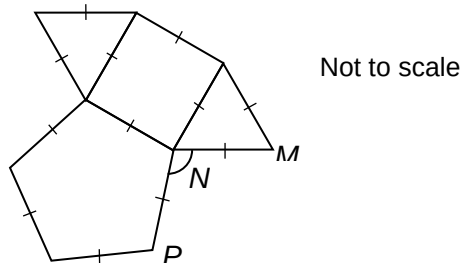
Skill: Students use relationships between angles in a triangle and on a straight line.

Additional questions

1. Given the information in the diagram, could line AB be parallel to line CE ?
2. If line AB was made to be parallel to line CE by moving point E , what would be the size of angle ECD ?

Question 4

A figure is composed of two equilateral triangles, a square and a regular pentagon.



What is the size of the external angle PNM ?

102°

☐

108°

☐

112°

☐

120°

☐

Answer: A

Skill: Students apply knowledge of angles in regular polygons.

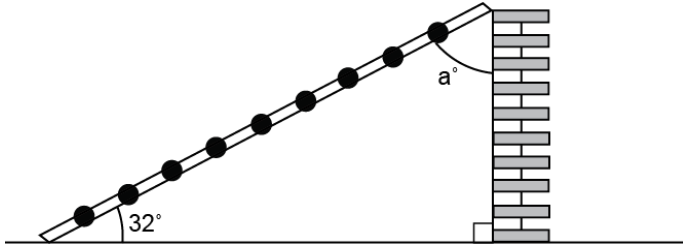
Additional questions

1. If the equilateral triangles are replaced with squares, what is the size of angle PNM ?
2. If the equilateral triangles are replaced with regular hexagons, what is the size of angle PNM ?
3. Assume the regular figures are floor tiles. Is it possible to replace the equilateral triangles with regular 20 sided polygons?

Student activity: Angle properties of familiar 2D shapes

Question 1

A ladder leans against a wall making an angle of 32° with the ground.

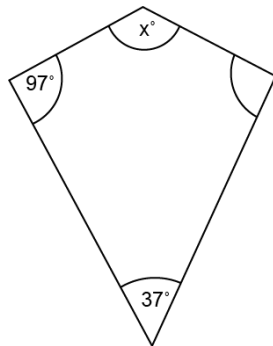


At what angle does the ladder meet the wall?

- ☐ 23°
☐ 48°
☐ 58°
☐ 78°

Question 2

A kite has angles as shown below.

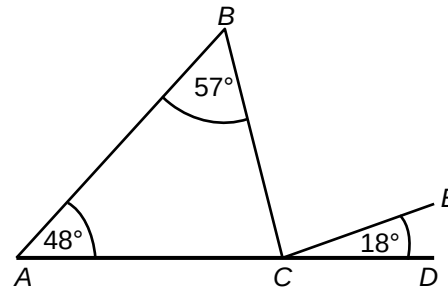


What is the size of angle x° ?

- ☐ 46°
☐ 97°
☐ 113°
☐ 129°

Question 3

In the diagram, ACD is a straight line.



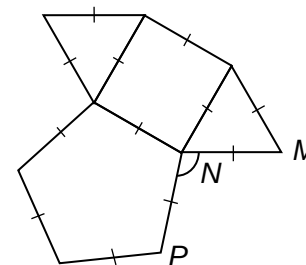
Not to scale

What is the size of angle BCE ?

- ☐ 18°
☐ 57°
☐ 75°
☐ 87°

Question 4

A figure is composed of two equilateral triangles, a square and a regular pentagon.



Not to scale

What is the size of the external angle PNM ?

- ☐ 102°
☐ 108°
☐ 112°
☐ 120°